Burns: Inhalation Injury Guidelines

Janine Koh, Joel Lim, Mok Yee Hui, Justin Wee, Grace Tan, Lynn Koh, Eileen Yap, Tan Wei Wei, Ong Chengsi

INTRODUCTION & DEFINITIONS

All burns patients should be assessed for inhalation injury. Fires/ explosions in **enclosed spaces** (e.g., house fires, car fires with prolonged extrication) convey the greatest risk for inhalation injury, which may be associated with burns to the face, neck and chest. It can also occur in the absence of cutaneous burns.

Inhalation injury comprises three main components:

1) **Thermal injury:** This is caused by direct thermal damage which mainly affects the upper airways. It can result in oedema, stridor and upper airway obstruction.
2) **Chemical injury to the airways and lung parenchyma (pneumonitis) from products of combustion:** This causes an intense inflammatory response and can lead to airway oedema, blistering, ulceration, erosions and sloughing of the mucosa. This local oedema and inflammatory response can result in bronchospasm, vasospasm, bronchorrhea, bronchial exudates and cast formation resulting in endoluminal obstruction. The debris in the airway is usually difficult to clear spontaneously due to the mucosal injury and disruption of mucociliary transport mechanism which predisposes to bacterial infection/pneumonia. Pulmonary oedema and ARDS may occur due to inflammatory capillary leak.
3) **Systemic poisoning from carbon monoxide poisoning and cyanide poisoning:** For enclosed fires, carbon monoxide (CO) and cyanide toxicities are an early and immediate cause of morbidity and mortality due to hypoxia.

ASSESSMENT

Administer high flow humidified oxygen via non-rebreather mask 15L/min O_2 to all patients.

Complete primary survey and thereafter complete secondary survey after initial resuscitation and once patient is stabilized. (*refer to Children's Emergency protocol for Paediatric Burns)

All children with suspected inhalational injuries (physical signs of soot in mouth/nose/larynx, hoarse voice, stridor, drooling, dysphagia, singed nasal hairs or carbonaceous sputum, wheezing, facial or neck burns) should be referred to ENT for prompt nasoendoscopy. Nasoendoscopy is useful to predict the need for intubation. The presence of upper airway oedema warrants elective intubation for airway protection (refer to Figure 1).

Patients should also be assessed for systemic poisoning from CO and cyanide poisoning. Blood carboxyhaemoglobin level should be obtained as soon as possible, while blood cyanide levels are not helpful due to a turnaround time of 3 days.

CHAPTER BURNS: INHALATION INJURY GUIDELINES

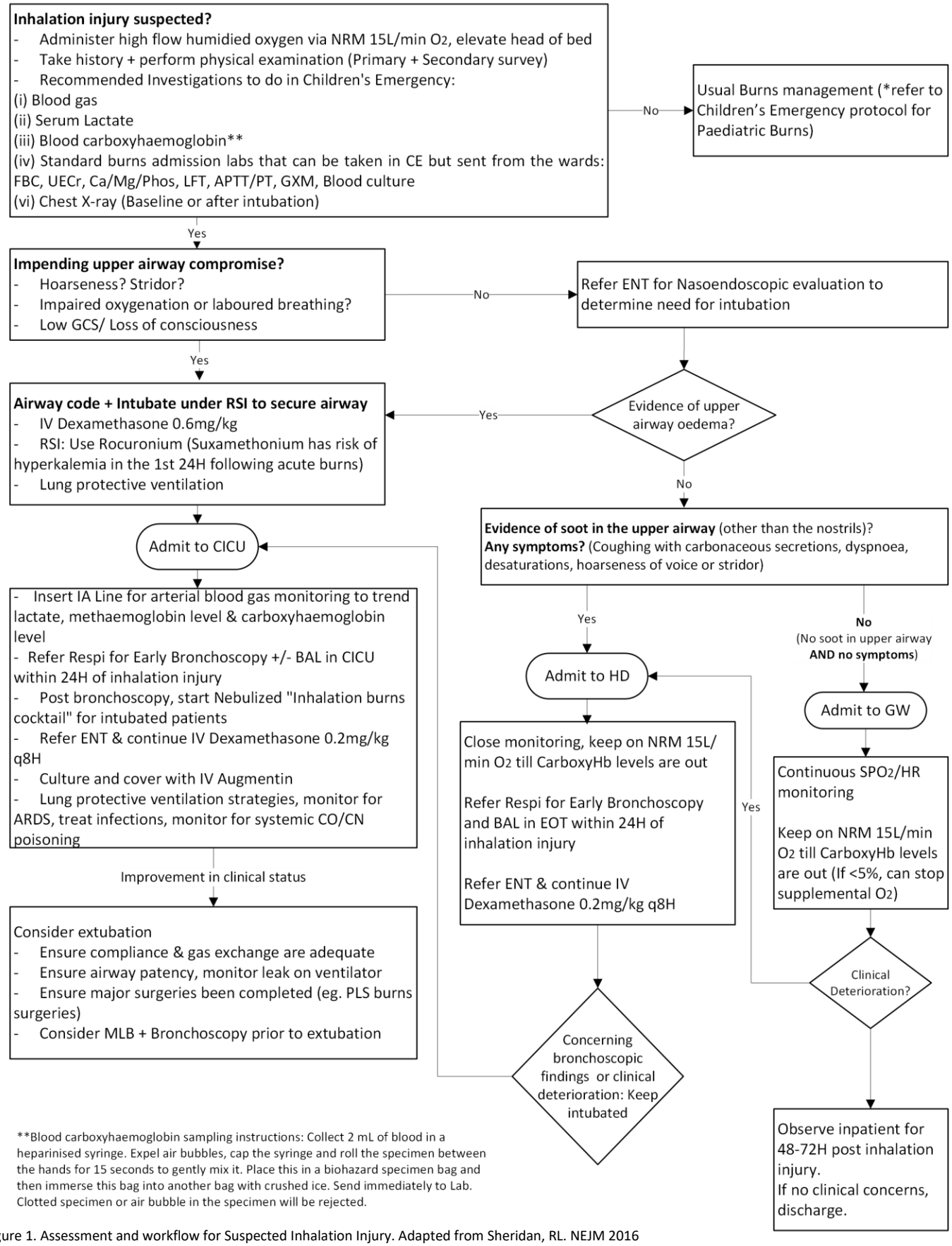


Figure 1. Assessment and workflow for Suspected Inhalation Injury. Adapted from Sheridan, RL. NEJM 2016

CHAPTER BURNS: INHALATION INJURY GUIDELINES

DIAGNOSIS

A) UPPER AIRWAY BURN INJURY

Early referral to ENT for nasoendoscopy to evaluate for carbonaceous debris in the pharynx or larynx, erythema or swelling in the pharynx, aryepiglottic folds and vocal cords.

Upper airway oedema develops rapidly over a few hours, particularly as fluid resuscitation is ongoing and may not peak until around 24 hours post injury. Therefore, early elective intubation for airway protection is important if there is evidence of upper airway oedema.

B) LOWER RESPIRATORY TRACT INJURY

The diagnosis of inhalation injury should be suspected based on clinical findings in the setting of prolonged smoke exposure; however definitive diagnosis is upon direct examination of the airways. Flexible bronchoscopy done within 24 hours of injury is the best diagnostic test for diagnosis of inhalation injury. However, we do not intubate for the sole purpose of performing a diagnostic bronchoscopy.

Bronchoscopic injury severity scoring such as the Abbreviated Injury Score (AIS) correlates with the degree of hypoxia experienced in the first 72 hours after injury and a higher AIS grade is associated with an increased risk of mortality, ARDS, longer duration of mechanical ventilation and overall length of ICU stay.

Flexible bronchoscopy is useful in patients with inhalation injury to evaluate the lower airways for involvement, and to clear soot particles (which can adsorb toxic substances that further prolong the inflammatory response, secretions, and casts. The AIS is the most frequently used score for bronchoscopic grading of inhalational injuries. AIS grade 2 and above correlates with mortality in inhalation injury. However, the intra- and inter-rater reliability of the AIS is not high, so photo or video recording is desirable, and clinical decisions should not be based solely on the AIS score. Changes related to inhalation injury may take up to 72 hours to develop, so repeat bronchoscopy 24-72 hours after the initial scope or pre-extubation can be considered depending on the severity of the initial scope findings and the clinical condition of the patient.

Injury to the lung parenchyma cannot be detected by bronchoscopy. The ventilation requirements of the patient should be assessed clinically. The P/F ratio (PaO_2/FiO_2) can help to assess parenchymal involvement, but it can also be affected by ventilator settings, volume of resuscitation used and metabolic poisons such as CO and cyanide. A P/F ratio < 300 mmHg following the initial fluid resuscitation (particularly at 36-72 hours post injury) has been found to be predictive of mortality in paediatric and adult patients. Initial CXR findings do not correlate well with clinical course or bronchoscopy findings, but may be useful as a baseline should there be further changes, e.g. pneumonia.

Abbreviated Injury Score (AIS) & Modified Endorf and Gamelli classification

Grade	Class	Examples	Mucosal appearance	Carbonaceous deposits	Bronchorrhea	Bronchial obstruction
0	No injury		No erythema or oedema	None	None	None
1	Mild injury		Minor or patchy area of erythema	Minor deposits in proximal or distal bronchi	None	None
2	Moderate injury		Moderate erythema	Moderate	Moderate	With or without compromise of the bronchi
3	Severe injury		Severe inflammation with friable mucosa	Copious	Copious	Bronchial obstruction
4	Massive injury		Severe inflammation; mucosal sloughing or necrosis	Copious	Copious	Endoluminal obliteration

C) CARBON MONOXIDE (CO) POISONING

Suspect CO poisoning in any patient who was exposed to a fire in an enclosed space. Prolonged smoke exposure causes elevated blood levels of CO that can lead to tissue hypoxia, profound hypoxic brain injury, respiratory failure, cardiac arrest and death. Pulse oximetry is unreliable in detecting CO poisoning and may be normal.

COHb Level (%)	Symptoms
0-5	Normal
15-20	Headache, Confusion
20-40	Disorientation, fatigue, nausea, visual disturbances
40-60	Hallucination, combativeness, coma, shock state
>60	Mortality >50%: Cardiopulmonary arrest and death

D) CYANIDE (CN) POISONING

Suspect Cyanide toxicity in fires in enclosed spaces (e.g., home, car) with synthetic materials (e.g., Plastic, vinyl/ PVC materials such as couches, car seats, mattresses, etc). Cyanide toxicity results in cellular/cytotoxic hypoxia by blockade of the mitochondrial respiratory chain.

Clinical Manifestations of Cyanide Poisoning	
Mild	**Moderate - Severe**
- Headache, dizziness, anxiety - Nausea, vomiting - Mild dyspnoea, tachypnoea and hypertension - Elevation in serum lactate (<8mmol/L) - High mixed venous oxygen saturation (low arterio-venous oxygen difference)	- Altered mental status (confusion, disorientation, coma) - Seizures, mydriasis - Bradypnea or apnoea - Hypotension or cardiovascular collapse - Severe metabolic acidosis with serum lactate ≥8mmol/L (lactate ≥10mmol/L is a sensitive and specific indicator of CN intoxication) - High mixed venous oxygen saturation (low arterio-venous oxygen difference)

Pulse oximetry is unreliable in detecting cyanide poisoning. Blood cyanide levels are not useful due to a turnaround time of 3 days. HSA can do expedited cyanide level testing on request, however the turnaround time is still 6-8 hours which may not be useful from a practical perspective. If cyanide poisoning is suspected, it should be treated empirically as risks associated with treatment are low.

TREATMENT

A) INHALATIONAL INJURY PROTOCOL FOR INTUBATED PATIENTS ("INHALATION BURNS COCKTAIL")

Nebulised N-acetylcysteine (NAC) in combination with aerosolized heparin has been proven to be effective in the treatment of inhalation injury by reducing ventilator days, risk of reintubation, atelectasis and mortality with no adverse bleeding events. NAC is a powerful mucolytic while inhaled heparin helps to prevent the formation of fibrin casts, which contribute to airway obstruction following inhalation injury. As NAC is an airway irritant and may produce bronchoconstriction, it is recommended to premedicate with bronchodilators. Bronchodilators have also been used in inhalation injury to decrease airflow resistance and improve dynamic compliance. Thus, the following cocktail of nebulized medications is recommended for all intubated patients with inhalation injury:

Cocktail of Nebulized Medications for Inhalation Injury Protocol	
Heparin 1000 units/ml	5ml (5000 units) every 4hourly
N-acetylcysteine 5g/25ml (20% NAC)	3ml (600 mg) every 4hourly
Premedicate with MDI salbutamol or Nebulized salbutamol to prevent bronchospasm	
Salbutamol	MDI 2 puffs or 0.03ml/kg/dose 5 minutes prior to Heparin and NAC nebulization

- Alternate heparin and N-acetylcysteine q2hrly for 7 days or until extubated as per the sequence below.
- Do not start on patients that are not intubated.

Sequence of Nebulization

Premedication (5mins before NAC)	**Time 0H**	**Premedication (5mins before Heparin)**	**Time 2H**	**Time 4H**
MDI/Neb Salbutamol	➔ Neb NAC	➔ MDI/Neb Salbutamol	➔ Neb Heparin	Repeat cycle

Example of the medication order

Heparin Sodium Injection - Other Route 5,000 unit, Q4H Nebulise Heparin 5ml (5000 units) every 4hourly Alternate Heparin nebulisation with N-acetylcysteine nebulisation
Acetylcysteine Vial/Ampoule - Inhalation 600 mg, Q4H Nebulise N-Acetylcysteine 3ml every 4hourly Alternate N-Acetylcysteine nebulisation with Heparin Nebulisation

B) CYANIDE POISONING ANTIDOTE

a. **Hydroxocobalamin 5g Powder (Cyanokit©)**

- First line, preferred antidote due to rapid onset, safety and efficacy

Dosing	Administration	Monitoring
IV 70mg/kg (max 5g), may repeat a second dose of 70mg/kg (max 5g) depending on severity and response Allow at least 30 minutes for response before considering second dose.[26] Maximum total recommended dose is 140mg/kg (or 10g), whichever is lower [Note: may also administer via Intraosseous route in children]	Preparation: Reconstitute the vial (5g) with 200ml normal saline (0.9%) (Concentration = 25mg/ml) Gently invert vial for 60s prior to infusion, do not shake. Resulting solution should be dark red in colour – if not, to discard. Administration: First dose: over 15min Second dose (if required): over 15min to 2 hours	Monitor for hypersensitivity reactions. Monitor blood pressure, heart rate during and after infusion. Monitor serum lactate levels and/or venous-arterial oxygen saturation gradient as a surrogate marker for response. If persistent lactic/ metabolic acidosis: assess for other causes of lactic/metabolic acidosis and discuss with toxicology expert regarding second dose. Suggest to check renal function daily for at least ≥7 days following therapy to ensure adequate clearance of various cobalamin-(III) complexes. A reversible red/pink colouring of the skin and mucus membranes and a dark red colouring of urine may occur. However, these effects are not clinically important.

b. Sodium thiosulfate 25% Injection (vial)

- Second line, less preferred as it has limited distribution into brain and limited penetration into mitochondria with slow onset of action
- In cases of massive CN ingestion or persistent toxicity, co-administration of hydroxycobalamin and sodium thiosulfate may be considered to provide potential synergistic effect.
- **Note**: Administer hydroxocobalamin first, followed by sodium thiosulfate. Do not administer Hydroxocobalamin and sodium thiosulfate simultaneously through the same IV line as they may form thiosulfate-cobalamin which is ineffective as an antidote.

Dosing	Administration	Monitoring
IV 300-500mg/kg (1.2-2ml/kg of a 25% solution); Maximum dose 12.5g (50ml of a 25% solution) over 10-20min	To be given undiluted over 10-20 minutes	Monitor the patient for 24-48 hours for adequacy of oxygenation and perfusion and for recurrent signs and symptoms of cyanide toxicity. If signs of cyanide poisoning are still present 30min to 2 hours after administration, half the original dose may be repeated.

C) MANAGEMENT OF CARBON MONOXIDE (CO) POISONING

The standard treatment for CO poisoning is FiO2 100% normobaric oxygen for 6 hours or until carboxyhaemoglobin (COHb) levels are <5%. The half-life of COHb is 1 hour when 100% FiO2 is administered compared to 5 hours on room air.

Hyperbaric oxygen for treatment of CO poisoning is controversial and has some clinical limitations. Wheezing and airway debris are relative contraindications to hyperbaric oxygen treatment as they increase the risk of gas embolism and pneumothorax at decompression.

Therefore, FiO2 100% normobaric oxygen is the safer and more practical treatment.

D) ANTIBIOTICS

- Consider empirical antibiotics for inhalation burns in patients who are mechanically ventilated or with signs of infection with clinical deterioration: IV Augmentin 40mg/kg/dose q8H (maximum 1.2g/dose)

Damage to the respiratory epithelium by chemical and thermal injury results in reduced mucociliary clearance which predisposes to bacterial infection. The incidence of pneumonia in patients with inhalation injury may be as high as 70% within a week of injury.

E) STEROIDS

- Steroids for airway oedema: IV Dexamethasone 0.6mg/kg STAT then 8hrs later 0.2mg/kg/dose q8H x 3 doses

MANAGEMENT CONSIDERATIONS

A) FLUID MANAGEMENT & NUTRITION

Fluid management

Fluid resuscitation should be used for children with cutaneous burns ≥15% TBSA using the modified Parkland formula. For children with cutaneous burns involving 10-14% TBSA, start IV maintenance fluids using the Holliday-Segar rule. *Refer to KKH Children's Emergency protocol for more details.

Be cautious not to over-resuscitate patients as over-resuscitation induces fluid creep with complications such as pulmonary oedema, pleural effusions, pericardial effusions, abdominal compartment syndrome, extremity compartment syndrome and conversion of burns to deeper wounds.

Nutrition

Early enteral nutrition within the first 24-48 hours of burn injury to optimize nutritional status is recommended. Enteral nutrition is preferred to parenteral nutrition as enteral nutrition nourishes the gut mucosa and reduces bacterial translocation, enhancing gut-associated immune function. **Refer all burns patients to the dietitian for feed and multivitamin recommendations.**

Feeds

Insert NG tube and aim to commence feeds within 6-8 hours of admission. Feeds can be commenced using the CICU feeding protocol using age-appropriate formula.

Multivitamin supplementation

In children with burn injury <20% TBSA, micronutrient requirements are similar to that of Daily Recommended Intakes (DRI) for age. A daily appropriate multivitamin can help to meet requirements (see table below). Children with burn injury ≥20%TBSA may require additional vitamin and trace mineral supplementation, particularly that of vitamin B1, C, D, E, copper, selenium and zinc.

Nutrient	**Inhalation injury + Minor burn (<20% TBSA)**	**Inhalation injury + Major burn (≥20% TBSA) Rodriguez 2011**	
Multivitamins	0-11months: 5ml Child Life multivitamins 1-12y: 10ml Child Life multivitamins ≥13y: 1 tab Multivitamin (Ocean's Health)		
Additional micronutrients (as necessary to meet recommended levels below)			
Vitamin A	Not necessary	0-13y: 2500-5000IU/d	≥13y: 10000IU/d
Vitamin C		0-13y: 250-500mg/d	≥13y: 1,000mg/d
Zinc		0-13y: 12.5-25mg/day	≥13y: 25-40mg/d
Copper		0-13y: 0.8-2.8mg/d	≥13y: 4.0mg/d
Selenium		0-13y: 60-140mcg/d	≥13y: 300-500mcg/d
Enteral formula	0-11 months: Expressed breast milk, donor human milk or infant formula 1-10 years: Pediasure or Nutren Junior >10 years: Ensure Life *If high gastric aspirates or feed intolerance, switch to hydrolyzed formula		

B) VENTILATION SETTINGS

The goal of mechanical ventilation should be to optimise oxygenation and ventilation while minimising potential ventilator-induced lung injury. Lung protective ventilation strategies (low tidal volume, high PEEP, limited plateau pressure, permissive hypercarbia) should be used especially in patients at risk of paediatric ARDS. In patients with ARDS who fail conventional mechanical ventilation, HFOV can be considered.

C) BRONCHIAL TOILET

Chest physiotherapy is recommended for patients with inhalation injury. The frequency of repeat bronchoscopies for bronchial toilet should be discussed with the respiratory team. Consider repeat bronchoscopy every 24-72 hours depending on initial scope findings and clinical situation, and pre-extubation scopes especially if extubated within the first 72 hours as changes related to inhalation injury may take some time to develop. Patients with clinical deterioration, more severe burns or large amounts of carbonaceous deposits in the airways should be prioritised.

D) EXTUBATION CRITERIA

- Ensure upper airway swelling has resolved by checking the endotracheal tube cuff leak.
- Successful ventilatory weaning and adequate oxygenation, spontaneous breathing trials and passed extubation readiness test (ERT).
- Good neurological status
- Improved respiratory secretions
- No further immediate surgical interventions planned
- Consider MLB + Bronchoscopy before extubation in discussion with ENT and Respiratory medicine

REFERENCES

1. Vanderbilt University Medical Center Pediatric Burn Inhalation Injury 2019 Protocol
2. Sheridan RL. Fire-Related Inhalation Injury. The New England journal of medicine. 2016;375(19):1905
3. Desai SR, Zeng D, Chong SJ. Airway management in inhalation injury: a case series. Singapore Med J. 2020;61(1):46-53. doi:10.11622/smedj.2019048
4. Goh SH, Tiah L, Lim HC, Ng EK. Disaster preparedness: Experience from a smoke inhalation mass casualty incident. Eur J Emerg Med. 2006;13(6):330-334. doi:10.1097/01.mej.0000224426.13574.b8
5. Ikonomidis C, Lang F, Radu A, Berger MM. Standardizing the diagnosis of inhalation injury using a descriptive score based on mucosal injury criteria. Burns. 2012;38:513–9. doi: 10.1016/j.burns.2011.11.009.
6. Bishop S, Maguire S. Anaesthesia and intensive care for major burns. Contin Educ Anaesth Crit Care Pain. 2012;12:118–22
7. Weaver LK, Howe S, Hopkins R, Chan KJ. Carboxyhemoglobin half-life in carbon monoxide-poisoned patients treated with 100% oxygen at atmospheric pressure. Chest. 2000;117(3):801-808.
8. Sen S. Pediatric inhalation injury. Burns Trauma. 2017;5:31. Published 2017 Nov 1. doi:10.1186/s41038-017-0097-5
9. Piantadosi CA. Diagnosis and treatment of carbon monoxide poisoning. Respir Care Clin N Am. 1999;5:183–202.
10. Demling RH. Smoke inhalation lung injury: an update. Eplasty. 2008;8:e27
11. Tibbles PM, Perrotta PL. Treatment of carbon monoxide poisoning: a critical review of human outcome studies comparing normobaric oxygen with hyperbaric oxygen. Ann Emerg Med. 1994;24:269–276. doi: 10.1016/S0196-0644(94)70141-5.
12. Buckley NA, Juurlink DN, Isbister G, Bennett MH, Lavonas EJ. Hyperbaric oxygen for carbon monoxide poisoning. The Cochrane database of systematic reviews. 2011:CD002041.
13. Sheridan RL, Shank ES. Hyperbaric oxygen treatment: a brief overview of a controversial topic. J Trauma 1999; 47: 426-35.
14. Lawson-Smith P, Jansen EC, Hyldegaard O. Cyanide intoxication as part of smoke inhalation--a review on diagnosis and treatment from the emergency perspective. Scand J Trauma Resusc Emerg Med. 2011;19:14. Published 2011 Mar 3. doi:10.1186/1757-7241-19-14
15. Goh CT, Jacobe S. Ventilation strategies in paediatric inhalation injury. Paediatr Respir Rev. 2016;20:3-9. doi:10.1016/j.prrv.2015.10.005
16. Oscier C, Emerson B, Handy JM. New perspectives on airway management in acutely burned patients. Anaesthesia 2014;69:105–10.
17. Endorf FW, Gamelli RL. Inhalation injury, pulmonary perturbations, and fluid resuscitation. J Burn Care Res. 2007;28(1):80-83.
18. Albright JM, Davis CS, Bird MD, et al. The acute pulmonary inflammatory response to the graded severity of smoke inhalation injury. Crit Care Med. 2012;40(4):1113-1121.
19. Mosier MJ, Pham TN, Park DR, et al. Predictive value of bronchoscopy in assessing the severity of inhalation injury. J Burn Care Res. 2012;33(1):65-73.
20. Charles WN, Collins D, Mandalia S, et al. Impact of inhalation injury on outcomes in critically ill burns patients: 12-year experience at a regional burns centre. Burns. 2022;48(6):1386-1395. doi:10.1016/j.burns.2021.11.018

21. Walker PF, Buehner MF, Wood LA, et al. Diagnosis and management of inhalation injury: an updated review. Crit Care. 2015;19:351. Published 2015 Oct 28. doi:10.1186/s13054-015-1077-4
22. Desai MH, Mlcak R, Richardson J, Nichols R, Herndon DN. Reduction in mortality in pediatric patients with inhalation injury with aerosolized heparin/N-acetylcystine [correction of acetylcystine] therapy. J Burn Care Rehabil. 1998;19:210–2.
23. Jeschke MG, Herndon DN. Burns in children: standard and new treatments. Lancet. 2014;383(9923):1168-1178. doi:10.1016/S0140-6736(13)61093-4
24. ISBI Practice Guidelines Committee; Steering Subcommittee; Advisory Subcommittee. ISBI Practice Guidelines for Burn Care. Burns. 2016;42(5):953-1021. doi:10.1016/j.burns.2016.05.013
25. Goh CT, Jacobe S. Ventilation strategies in paediatric inhalation injury. Paediatr Respir Rev. 2016;20:3-9. doi:10.1016/j.prrv.2015.10.005
26. Queensland Poisons Information Centre. https://www.poisonsinfo.health.qld.gov.au/for-health-professionals/antidote-stocking-recommendations/hydroxocobalamin
27. Avni T, Levcovich A, Ad-El DD, Leibovici L, Paul M. Prophylactic antibiotics for burns patients: systematic review and meta-analysis. BMJ. 2010 Feb 15;340:c241.
28. Bai C, Huang H, Yao X, Zhu S, Li B, Hang J, Zhang W, Zarogoulidis P, Gschwendtner A, Zarogoulidis K, Li Q, Simoff M. Application of flexible bronchoscopy in inhalation lung injury. Diagn Pathol. 2013 Oct 21;8:174.
29. Kumana ET, Charles WN, Milton-Jones H, Agbontaen K, Soussi S, Dunn K, Davies R, Atkins J, Beynon C, Charbonney E, Gantner D, Giles J, Jones I, Martin N, Pantet O, Shelley O, Sisson A, Sokhi J, Stewart BT, Vorster T, Vizcaychipi MP, Wood F, Yarrow J, Singh S; International Burns Inhalation Injury Consortium∗. Evaluating inter-and intra-rater reliability in the bronchoscopic grading of burn inhalation injury: The iBRONCH-BII study. Burns. 2025 Jun;51(5):107502.